

IT

Lec 1

Why Learn about Computers?

Exchange e-mail

Shopping

In the home

look up information

Smart homes

work-related tasks

Normal coursework

In Education

To facilitate

distance learning

Research tools

For Scientists

In all businesses

Electronics manufacturing

By military personnel

productivity tools

for office workers

Internet applications

online retailing

POS systems

On the Go

In Communication

Social network sites

What does Computer do? How it works?

① Input → entering data into the computer.

② Processing the data.

③ Output → Presenting the results.

④ Storage → Saving data, programs, or for future use.

Data & Information

Data Processing into a meaningful form → Information

Information processing: Conversion of data into information.

Generations of Computer

	First generation 1946 to 1957	Second generation 1958 to 1963	Third generation 1964 to 1970	Fourth generation 1971 to present
Hardware	Vacuum tubes & Magnetic drums	Using Transistor	(IC) chips	(LSI) & (VLSI)
Software	In machine language	In assembly language	High-level languages	(GUI) & High- level languages
Computation Time	In millisecond	In Microseconds	In Nanoseconds	In Pico seconds
Physical Appearance	Enormous in size	The size was reduced	Quite small	Smaller than computers of the previous Generation
Application	For Scientific applications	High cost of commercial production	Produced commercially	Available commercial & personal
Examples	ENIAC & UNIVAC	PDP IBM 1401 CDC 1604	IBM 370 PDP 11	IBM → 1st one to home use Apple → Macintosh

Use SLSI chips

Have large memory requirements

Fifth generation

"present & next"

Based on (AI)

Try to simulate the human thinking

programming languages

Machine language

Assembly language

High-level language

Characteristics of Computers

- ① Speed
- ② High degree of accuracy
- ③ Doesn't get tired
- ④ Versatility
- ⑤ Storing large volumes of data & information.

The most Powerful & expensive computer

- So Fast
- Used in sending astronauts into space, controlling satellites, forecasting the weather, exploring for oil & breaking codes

Powerful computer & expensive used by hospitals, Banks, universities & Government offices

Tiny embedded computer perform specific tasks

Mini Computer used to host data for a small network

Supercomputer Mainframe

Embedded computers

Computers

Servers

PCs

Mobile devices

Small device has built-in computing or internet capability ex. smartphone & tablet

Portable computers

- Easy carried
- Flexible
- Rechargeable batteries

Notebook computers

Tablet computers

Hybrid notebook-tablet c.

Netbooks

Desktops

Mac

PC-compatible

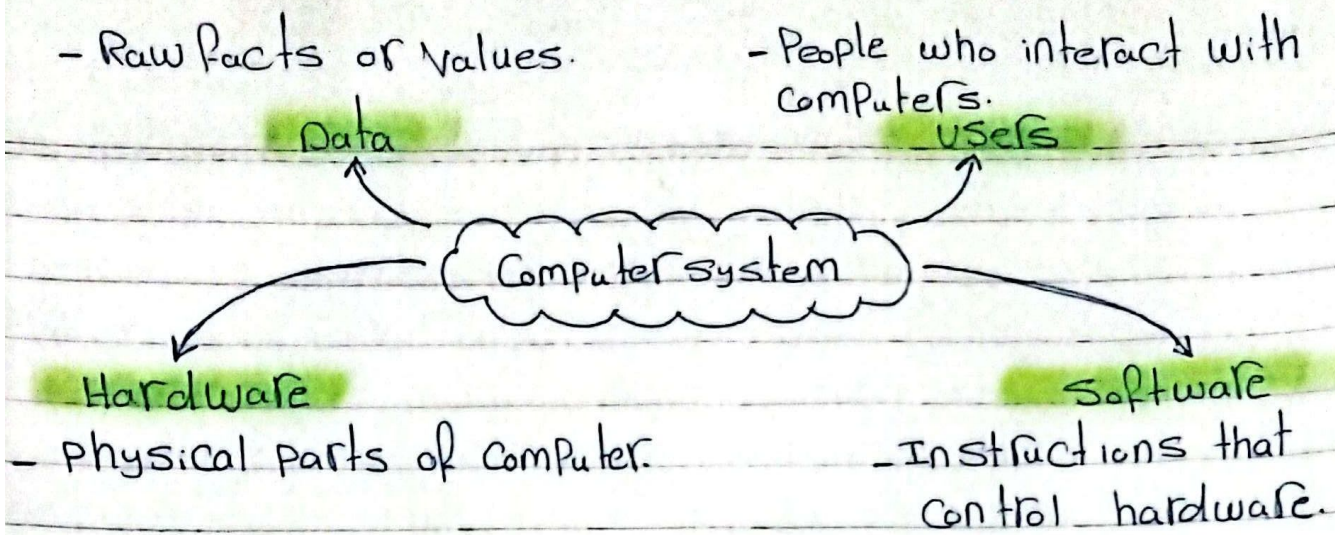
Mac OS

Microsoft windows system

- * Powerful desktops called workstations
- * Mainframe more powerful than servers.

Convertible

Limited features



Computer Hardware

- Memory unit → stores data & instructions.
- Arithmetic logic unit (ALU) → Performs logic operations.
- Input unit → Moves data from out into the computer.
- Output unit → Displays results to users.
- Control unit → The stage manager to insure the all components act in concert.

Digital Data Representation

- Computers process binary data (0s and 1s)
- Text representation → ASCII, EBCDIC & unicode
- Graphics → Bitmap images (Pixel-based)
- Audio → Converted from analog to digital.
- Video → Sequence of still frames creating motion.

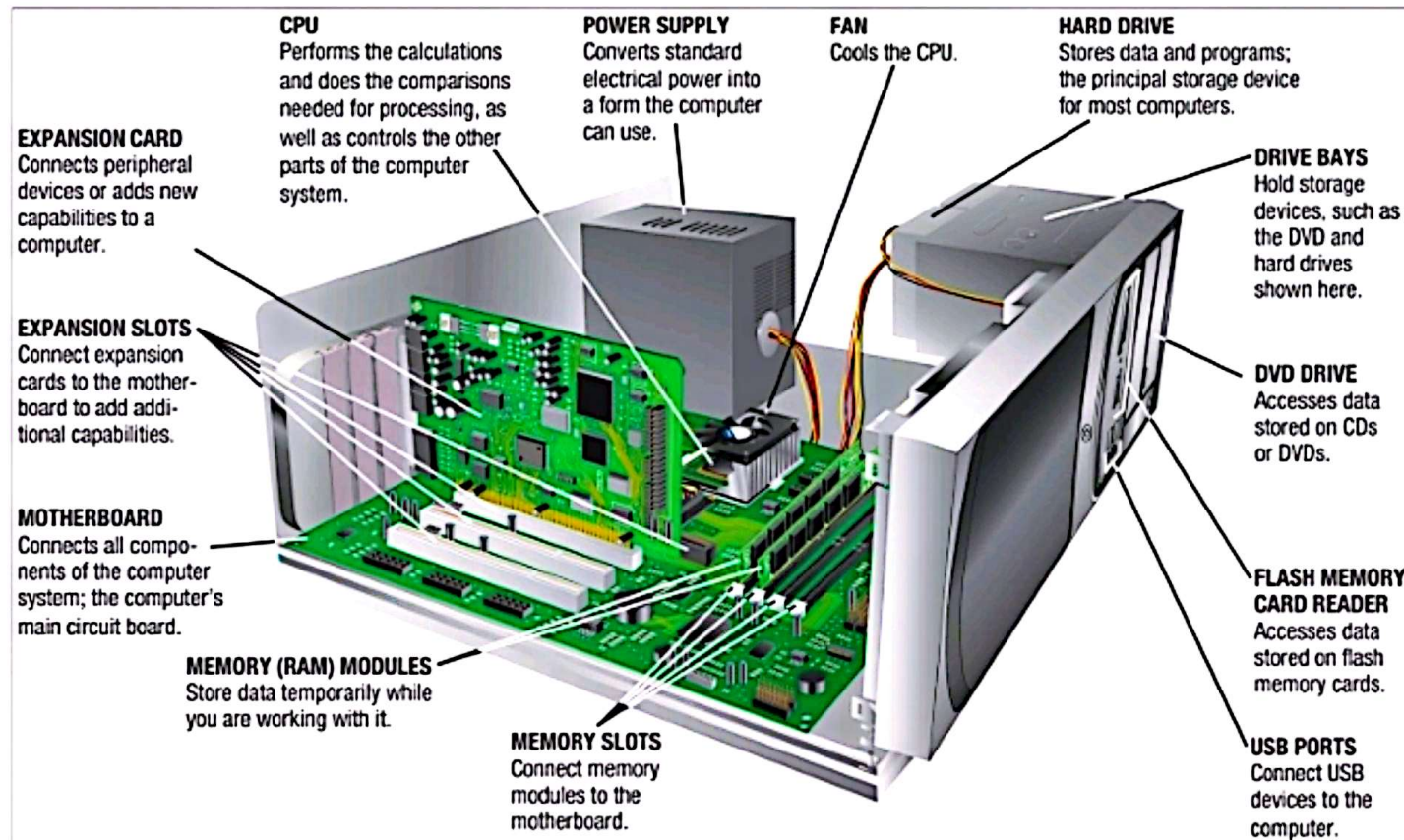
Inside the system unit

- Motherboard → Main circuit board connecting all components.
- Power supply → Converts electricity for the system.

Processors

- CPU → The brain of the computer.
- GPU → Handles graphics processing.
- * Processing speed is measured in GHz.

3.1 INSIDE THE SYSTEM UNIT



1. Digital Data Representation

- All Computers digital Computer (Binary Computer)
- All data Processed by binary $\rightarrow$ (0s & 1s) $\rightarrow$ bits
- Large number bits needed written, computer Program, digital photo
- 8 bits together are collectively referred to as $\rightarrow$ byte

1. Representing Numerical Data

Decimal numbering system

- Uses 10 symbols
- 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
- $\rightarrow$ all possible numbers

7216 $\rightarrow$

10^3	10^2	10^1	10^0
7	2	1	6

Binary numbering system

- Uses Two symbols
- 0, 1 $\rightarrow$ all possible number & perform math computation

1001 $\rightarrow$

2^3	2^2	2^1	2^0
1	0	0	1

* Position each digit determines The Power, or exponent
The base number (10 For decimal or 2 For binary) is raised

2. Coding Systems For Text Based

ASCII, EBCDIC

Character sets

- unique combination of 8 bit (1 byte)
- allow 256 (2^8 = up to 256 characters)

Unicode

- represent text-based data written any ancient or modern language
- longer code consisting 1 to 4 bytes (8 to 32 bit) can over one million characters

3. Coding systems for graphics, Audio & video ✓

- Graphics → Most common method for storing form of bitmap image called Pixels
- Audio → must be in digital form to be stored on a storage medium
To convert analog sound to digital sound
- Video → using a collection of frames each frame contains a still image
illusion of movement is created
must be convert instruction into binary code

2. Computer Hardware ✓

3. Inside the system unit ✓

3.1. The motherboard

- circuit board is thin board containing computer chip and electronic component
 - need be connected via a wired or wireless connection
- The motherboard

3.2. The Power Supply

connects to the motherboard to deliver electricity to the computer

3.3. Processors

→ Speed

- The Primary Processor is Central Processing Unit (CPU)
called → microprocessor
 - servers today use Intel or Advanced Micro Devices (AMD)
 - Graphics processing unit (GPU)
↳ take the processing, quality, pure image
- Processor → measure GHz or MHz

word size

measure in bits or bytes, word size allows for faster processing

Bus width, Bus speed and Bandwidth

Bus is electronic path over data can travel

↳ Inside the CPU & motherboard

number bits dependent on bus width

bus speeds also very important factor because the bus width and bus speed together determine the bus's

bandwidth → the amount of data that can be transferred via the bus in each time period

Memory

Random access memory or RAM is located inside the system unit, used by store data on short-term

Long-term storage → internal hard drive or movable storage media

Four other types computer memory

1. Random Access Memory (RAM)

also called main memory or system memory used essential part of the operation system

measure → bytes

must be able to find data & programs

2. Cache Memory

very fast memory, used to speed up processing

3. Registers

high-speed memory, increase CPU performance

4. Read-only Memory (ROM)

Permanently store data (firmware)

chips are attached to the motherboard

cannot write over the data

replaced with flash memory.